

Evaluation of a collaborative reading annotation system through multimodal data analysis

Lili, Liu*

Faculty of Artificial Intelligence in
Education, Central China Normal
University, Wuhan, People's Republic
of China

lliliu@mails.ccnu.edu.cn

Lishan, Zhang†

Faculty of Artificial Intelligence in
Education, Central China Normal
University, Wuhan, People's Republic
of China

lishan.zhang@ccnu.edu.cn

Yan, Shu

Faculty of Artificial Intelligence in
Education, Central China Normal
University, Wuhan, People's Republic
of China

2021114179sy@mails.ccnu.edu.cn

Shuwen, Wang

Faculty of Artificial Intelligence in
Education, Central China Normal
University, Wuhan, People's Republic
of China

shuwenwang94@163.com

Han, Wu

Faculty of Artificial Intelligence in
Education, Central China Normal
University, Wuhan, People's Republic
of China

hanwu289@163.com

Min, Xv

School of Chinese Language and
Literature, Central China Normal
University, Wuhan, People's Republic
of China

xvmin2022@163.com

ABSTRACT

Technological advances and the explosion of epidemics have contributed to a surge in the number of online learning platforms. Because single modal data is often not enough in evaluating the usability of interface interaction design for online learning platforms and multimodal data (eye movement, Electroencephalogram, skin conductance response) with advanced sensing technologies provide new possibilities to address this issue, this case study explores how multimodal data can be used to evaluate the interface design for our self-developed collaborative reading system. The results of our randomized between-subject experiments showed that, from eye movement analysis, constructive-level annotations prompt students to allocate more attention to the annotation area than active level annotations and facilitate the transition between the annotation and reading areas. From EEG data analysis, all the students stayed high concentration levels no matter the types of annotations they were reading. From SCRs analysis, although no significant difference in the level of excitement between experimental conditions was identified, students showed great individual difference within the same conditions. This study illustrated how multimodal data can be applied to interface design evaluation.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

*Share the first authorship

†Share the first authorship

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3573854>

KEYWORDS

Collaborative reading annotation system, Multimodal data analysis, Machine learning

ACM Reference Format:

Lili, Liu, Lishan, Zhang, Yan, Shu, Shuwen, Wang, Han, Wu, and Min, Xv. 2023. Evaluation of a collaborative reading annotation system through multimodal data analysis. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3573854>

1 INTRODUCTION

With the development of intelligent technology and the impact of the COVID-19 epidemic, online learning has gained more attention than ever before. This brought the explosive growth of online learning platforms, for example, online collaborative reading system. The system provides easy accesses to online teaching, teacher-student interaction, and student-student collaboration. To design such a system, interface layout and user acceptance are usually considered as essential factors. The latest results inferred from self-report questionnaires and eye-tracking technology noted that, vertical layout panel design was more efficient for improving user learning efficiency than horizontal layout panel design [1]. Christensen [2] indicated that the individual's acceptance was also needed to be considered during system design, because it can enhance user participation. From learners' perspectives, it is important to easy and fast access to their target knowledge while using online learning platforms. From instructors' perspectives, it is important to conduct online teaching conveniently. Therefore, for designers, it is crucial to develop a user-friendly interface considering the needs from both learners and instructors. Comprehensive evaluation methodologies are required to evaluate interface layout and user acceptance. Although the traditional methods such as user study questionnaire have been widely used and accepted by HCI community [3], they are solely based on subjective evaluation which may lead to unexpected bias. This type of single modal data may not be enough to evaluate the usability of interface interaction design. In contrast,

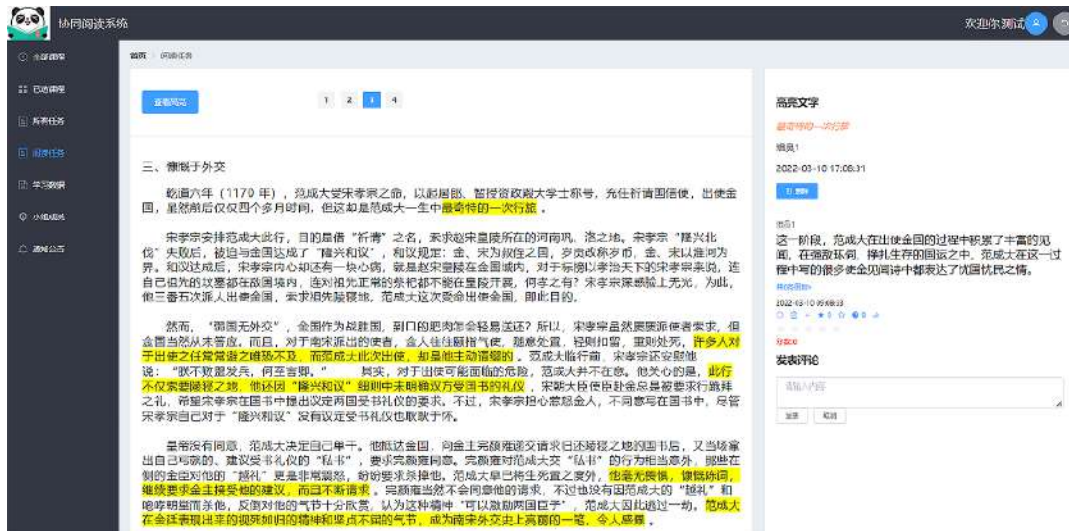


Figure 1: Collaborative reading system.

the multimodal data such as eye movement, Electroencephalogram and skin conductance responses offer a new way for evaluation from mental and physical perspectives [4-7]. Multimodal data analysis can reveal the participants' cognitive process and emotional state, which is a promising way to assist in interface design [7, 8].

Different sensing devices perceive different types of physiological data. Eye tracker can capture the position of fixation, vision shifting between areas of interests (AOIs), and changes of pupil diameters. Based on these representations, it is possible to model eye movement patterns. These patterns can reveal the influences of interface layout on learning from a cognitive perspective [9]. Brainwave device can safely acquire and output EEG power spectra (e.g., alpha waves, beta waves) and attention values. This modal can reflect the participants' immersion and concentration levels of their mental states [10]. The higher the value is, the more concentrated the participant is; conversely, the lower the value is, the more distracted the participant is [11]. The device such as Empatica E4 can measure skin conductance response, heart rate, and Blood Volume Pulse. This modal can reflect the participants' excitement levels from emotional perception [12].

We developed a collaborative reading system that supports highlighting, annotating, and peers' annotations displaying. The system's interfaces include login, course selection, learning task selection, and an annotated page. This case study aimed to explore the influences of the interface of the collaborative reading system on learning. Multimodal data (eye movement, EEG, skin conductance responses) analysis with self-report questionnaire together can provide a comprehensive evaluation.

2 METHODS

2.1 Participants

We recruited a total of 48 students to participate in the experiment. However, one participant's sample was excluded due to its eye movement sampling rate below 60% [13]. The reported results

are based on the remaining 47 participants (13 males, 34 females; ages ranged from 18 to 27 years: $M=22.426$, $SD=2.204$). Specifically, these participants majored in different disciplines including Computer Science and Technology, Education Technology, Psychology, Education, Artificial Intelligence, and Information Management. All participants had normal or corrected visual acuity, no color blindness, and no astigmatism. They were randomly assigned to the experimental group (24 students) and the control group (23 students).

2.2 The system

There are two core areas on the annotated page: a readable area and the area to annotate. The readable area enables participants to highlight what they think is essential, then the participants can make annotations associated with the highlighted text. In this case study, we set up three kinds of highlighting permissions: visible only to the users themselves, visible only to the group members, and publicly visible. By default, the permission to highlight text is visible only to the users. Similarly, the permission to annotate is tied to the permission to highlight. The annotated page of the collaborative reading system is shown in Figure 1.

2.3 Materials

In the fall semester of 2021, the collaborative reading system was applied to the Poetry Garden course at one of the universities in China, and most students made at least one annotation on this platform while reading. Two researchers coded these annotations generated by students in the course based on the interactive, constructive, active, and passive (ICAP) framework [14-16]. Then, we proposed two cognitive levels of annotations (i.e. constructive level and active level) based on the coding results. To further explore how these two different types of annotations influence the readers, we discussed and designated 13 phrases as highlighted texts, and assigned each highlighted text two types of annotations (i.e.

active and constructive). For example, the highlighted text is "Worrying about the crops and pitying the old peasants". "It showed Fan Chengda's love and hatred, his disgust for the corrupt official culture, and his virtue of being an officer who cares for the people" is considered as a constructive annotation, and "Worrying about the crops and the poor old men" is considered as an active annotation. The constructive annotations were visible to the participants only in the constructive condition (i.e. the experimental group), while the active annotations (i.e. the control group) were visible to the participants only in the active condition.

2.4 Instruments

2.4.1 Eye tracker. The eye movement data was collected using a Tobii Pro X3-120 eye tracker, developed by Tobii Technology in Sweden, with a sampling rate of 120 Hz and an accuracy of 0.4°. The eye-tracking device was connected to a 24-inch monitor with a resolution of 1920 × 1080 pixels and collected data automatically. The Pro Lab software that accompanies the eye-tracking device is used to analyze and export data.

2.4.2 Electroencephalograph (EEG). The EEG data was collected by a NeuroSky brainwave device with a sampling rate of 1 Hz, which can safely acquire and output EEG power spectra (e.g., alpha waves, beta waves), eSense concentration index, eSense relaxation index, and blink. The single-channel device consists of a headset, an ear clip, and a sensor support arm. The reference and ground electrodes are on the ear clip. And the EEG sensor is on the sensor support arm and attached to the forehead above the eye at FP1.

2.4.3 Skin conductance responses (SCRs) wristband. The physiological activities were detected using wearable sensors Empatica E4. It is the only wearable on the market to combine EDA sensor with a sampling rate of 4Hz and PPG sensor with a sampling rate of 64Hz. In particular, EDA sensor is often used to measure skin conductance responses. And PPG sensor can be usually used to measure Blood Volume Pulse, from which heart rate variability can be derived.

2.5 Procedures

We conducted experiments based on a virtual online collaborative reading scenario. All participants were randomly assigned to the experimental and control group. Before the experiment, all students read the purpose and procedure of the experiment, signed an informed consent form, and completed the basic questionnaire about demographic information such as name, major, and grade. The experimental procedures were as follows: a) Before the start of the experiment, each participant completed the pretest, which was used to examine their prior knowledge level; b) The participants were seated at a distance of 60-80 cm from the eye tracker and wore the EEG device as well as the E4 wristband; c) The participants fixed their heads on a neck rest and conducted a 9-point eye movement calibration; d) Each participant finished reading the article within 15 minutes; e) After the experiment, the participants completed the posttest and received the compensation. A few participants volunteered to take a semi-structured interview.

The experimental procedures are shown in Figure 2. Before the formal experiment, 10 participants took part in a pilot study. On

the one hand, we checked for possible problems about the experimental material, the pretest, and the posttest. On the other hand, we tried to estimate the overall reading time of the material. Based on the results of the pilot study, we adjusted the content of the questionnaires and set 15 minutes for reading the experimental material. The experiment setting is shown in Figure 3.

3 ANALYTICS

3.1 Knowledge and psychological test

There was no significant difference between the experimental and control groups in the pretest ($t=1.587$, $p=0.119$), which indicated that the two groups were approximately at the same level of initial knowledge level. After learning through reading, there was still no significant difference between the two groups in the posttest ($t=1.205$, $p=0.234$). However, there was a borderline significant difference ($t=1.888$, $p=0.066$) in the open-response questions of the posttest. Specifically, the difference was sourced from the analysis of the poet's experience ($t=2.041$, $p=0.047$), and the experimental group students ($M=6.125$, $SD=3.575$) scored significantly higher than the control group students ($M=4.098$, $SD=3.216$). There was no difference in the other four subdimensions (the meaning, background, details and emotion of poem). Learning gain (LG) was computed as well to confirm the findings. LG was calculated as the posttest score minus the pretest score. The difference in LG was sourced from the analysis of the poet's experience ($t=2.081$, $p=0.043$), and the experimental group students ($M=5.802$, $SD=3.573$) scored significantly higher than the control group students ($M=3.761$, $SD=3.124$). Participant self-reported their perceived learning in the posttest. And a significant difference between the two groups was found ($t=2.200$, $p=0.033$). The experimental group ($M=6.080$, $SD=0.974$) perceived significantly higher learning than the control group ($M=5.348$, $SD=1.301$). There was no significant difference in other cognitive aspects. The results are summarized in Table 1.

3.2 Eye movement

Eye movement patterns can reveal the influences of interface layout on learning from a cognitive perspective [9]. Unlike the previous test analysis, machine learning technologies (sequential pattern mining and association rule analysis) were used to illustrate the influences of interface layout on visual attention. This case study has three areas of interest (AOIs), including the highlighting area, the nonhighlighting area, and the annotation area. Before analysis, eye movement coordinates were replaced with the AOIs. Thirty-four behavioral patterns were obtained for the experimental group, and 33 behavioral patterns were obtained for the control group. The top 20 subsequences with the highest frequency in the two groups were selected as the sequences to be further analyzed. We focus on the sequence patterns that reflect how students switch between the highlighting and annotation areas. The sequence patterns of "Annotation->Highlighting" and "Highlighting->Annotation" appeared 237 and 245 times in the experimental group, and 170 and 148 times in the control group. The frequency of these two patterns in the experimental group was higher than that in the control group. Sequences with a frequency of more than 50 were selected for the association rule analysis. Unlike the normal association rule,

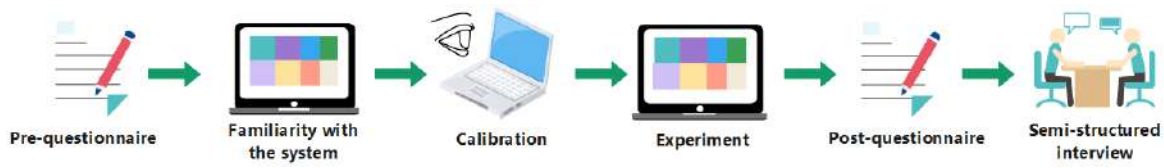


Figure 2: Experimental procedure

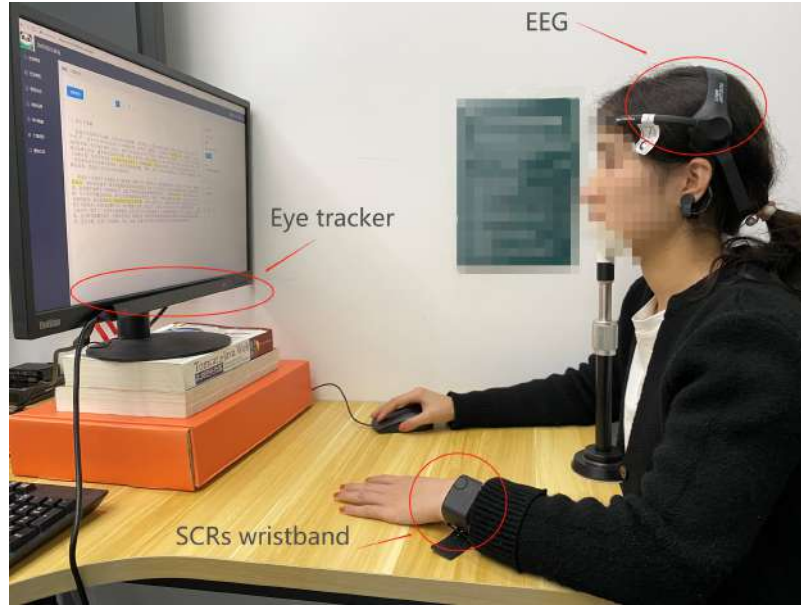


Figure 3: The setting of the experiment

Table 1: Knowledge and psychology test

Metric	Experimental group		Control group		t	p	Cohen's d
	M	SD	M	SD			
Pretest scores	15.396	3.119	13.902	3.332	1.587	0.119	0.463
Posttest scores	63.048	11.252	59.485	8.807	1.205	0.234	0.352
Open-response question scores	26.865	9.062	22.250	7.596	1.888	0.066	0.551
Learning outcomes	11.469	9.492	8.348	6.721	1.296	0.202	0.378
Learning gain	6.080	0.974	5.348	1.301	2.200	0.033*	0.642
Perceived learning	3.630	0.970	3.650	1.191	0.086	0.932	0.025
Intrinsic cognitive load	5.250	1.225	5.830	0.937	1.806	0.078	0.527
Germane cognitive load	3.170	1.685	3.700	1.987	0.986	0.330	0.288
Extrinsic cognitive load	5.420	1.316	5.000	1.128	1.163	0.251	0.339
Interest in learning	22.92	3.810	23.65	3.157	0.719	0.476	0.210
Total learning inputs	5.67	1.494	5.040	1.331	1.508	0.139	0.440
Psychological assessment	2.130	1.191	2.090	1.041	0.116	0.908	0.034
Impact of annotation							
Positive/Negative							

* $p < 0.05$, ** $p < 0.01$

the order of occurrence of the elements in the sequence was considered here. The confidence level of the sequence "(Highlighting, Annotation) -> Highlighting" occurring was 40% in the experimental

group and 21% in the control group. The confidence level of the sequence "(Annotation, Highlighting) -> Annotation" was 2%, and the confidence level of the sequence "(Nonhighlighting, Highlighting,

Annotation)->Highlighting" was 16% in the experimental group but not in the control group.

3.3 EEG

EEG data can reflect the participants' immersion and concentration while learning from their mental state [10]. In this study, each participant's attention was measured by the NeuroSky brainwave device. The attention ranged from 0 to 100. The higher the value is, the more concentrated the participant is; conversely, the lower the value is, the more distracted the participant is [11]. The baseline for this study was the mean attention value for 10 seconds after the participant logged into the system. The times participants looked at the comments area was counted, and the rate above the baseline was calculated. For example, in the experimental group, the total attentional time of participant A is 102 s, and the baseline is 39.4, of which 96 s are above the baseline level, and the rate of attentional periods above the baseline is 0.941176. According to the results of the independent samples t-test ($t=0.841$, $p=0.405$), there was no significant difference between the experimental group ($M=0.551$, $SD=0.251$) and the control group ($M=0.489$, $SD=0.244$). The non-significant difference could be that the attention level was generally high in both groups during the experiment. That also shows that the differences in perceived learning are not caused by the degree of engagement.

3.4 Skin conductance responses (SCRs)

Skin conductance responses reflect the participants' excitement levels from emotional perception [12]. In this case study, the SCRs measured by the EDA sensor were selected as the analysis indicators. EDA signals were processed using the LEDALAB toolbox [17]. The signal was decomposed into the tonic skin conductance level and the transient skin conductance response using the continuous decomposition analysis [18]. The mean value of EDA data for the first 20 seconds of recorded data was selected as the baseline. T-tests were performed on the number of SCR peaks. According to the results of the peaks t-test ($t=1.597$, $p=0.121$), there was no significant difference between the experimental group ($M=12.33$, $SD=15.466$) and the control group ($M=25.52$, $SD=36.593$). In addition, the big standard deviation implied high individual differences.

4 DISCUSSION

Previous students' everyday usage of the collaborative learning system can generate valuable learning interventions for future students. Learners can comment on a particular highlighted text, discuss it in the chat room, or give him/her a thumbs-up/thumbs-down when the highlighted text is public visible. Peers can express their own opinions rather than just immerse themselves in reading. Because constructive-level annotations can improve learners' performance, if previous students' annotations can be filtered by algorithms for retaining constructive-level annotations, it will benefit future students' learning. In addition, the collaborative learning activities supported by technology may help foster individual active learning and critical thinking [19, 20] because online collaborative learning stimulates innovation and creativity [21].

Constructive-level peer annotations facilitate learning outcomes. The experimental group students perceived more learning

than the control group according to the psychological questionnaire. Specifically, this distinction was mainly sourced from analyzing the poet's experience of the two open-response questions. These results showed that students could acquire more knowledge under the constructive-level annotations condition. However, the learning behavior data from the log file showed no significant difference in their first-round reading between the experiment and control groups. Indicators from the eye tracker further showed that participants under the constructive-level annotations condition allocated more attention to the annotation area. These results showed that constructive-level annotations can attract learners' attention and facilitate learning performance. In recent years, an interesting line of research concerning the quality of annotations has investigated how seed annotations may improve the effectiveness of social annotations. Miller, et al. [22] took high-quality annotations from an online course of the previous academic year, randomly selected half of the study sections in the new course round, and posted these annotations before class as seed annotations. The results suggested that the seeded sessions elicited longer and higher-quality discussion threads. Based on this case study, if high-quality annotations are filtered using text classification algorithms such as machine learning and then embedded into the online collaborative reading system, this initiative will benefit reading instruction and students' learning. This is also what we will strive for in the future.

Multimodal data can be used to comprehensively evaluate interface design. According to the results above, single modal data analysis (EEG or SRCs) could not explain interface interaction design well. However, the eye movement patterns provide strong evidences for the effects caused by the differences of interface designs (content of the annotation area: constructive/active level). Especially in the laboratory environment, multimodal data provides a possibility to fully evaluate interface design usability. Yan, et al. [23] used a combination of eye movement (including fixation duration, number of fixations, and single gaze duration), EEG (including α wave and β wave), and behavioral response (emphasize rapid visual orientation for accurate information acquisition) methods to assess the optimal range of icon and display resolution in interface design. Jiang, et al. [24] used performance (task time), self-report (e.g., task difficulty and workload), eye movement (e.g., pupil diameter, total gaze time), and physiological indicators (including skin conductance response and skin temperature) to assess each participant's perceptions of the usability of radiotherapy software. The research results noted that the use of multimodal measures provided a more comprehensive assessment of participants' intuitive perception of the usability of the radiotherapy software.

The suggestions for improving the interface design from the participants. A few subjects volunteered to participate in a semi-structured interview after the experiment. The semi-structured interviews focused on the usability, rationality, and user-friendliness of the overall layout of the interface (e.g., the layout of the navigation, reading, and annotation areas facilitated/hindered learners' learning) and functional settings (e.g., the impact of highlighting permissions and colors, the font and local layout of the annotation area on learners). One participant suggested, "I would like to have a variety of colors to choose when highlighting text." Similarly, another participant said, "When multiple people highlight the same text, it will be difficult for me to distinguish myself from others as

the color is all yellow." Adding various text highlighting colors for users to choose from is part of our current work. In addition, one participant mentioned that "I would like to change the permissions for annotations. For example, I had set some annotations to be visible only to the group, but now I want to change them to be visible only to myself." However, one subject considered, "These highlighted texts would prevent me from reading. For example, my mind would be interrupted." These suggestions not only provide new directions for our future work but also have implications for other similar aided instruction platforms supporting collaborative reading. In the future study, we will continue optimizing the interface design, implementing the additional function, and regularly evaluating the interface to support teaching and learning.

5 CONCLUSION

The collaborative reading system enables students to learn collaboratively, and each learner can gain more engagement by highlighting, annotating, discussing. Under the constructive-level annotations condition, learners can perceive more knowledge and achieve better academic performance. The eye movement sequence mining and association rule analysis further explain the results. Constructive-level annotations prompt participants to allocate more attention to the annotation area and facilitate the shifting of vision between the annotation and reading areas to some extent. Furthermore, EEG analysis showed that participants stay high levels of concentration no matter which condition they are at. From SCRs analysis, although there is no significant difference at the level of excitement under different conditions of annotations, there is a great deal of individual difference within the same condition.

This case study has the following contributions: a) Collaborative reading annotation system can generate valuable learning interventions for future students; b) Constructive-level annotations in the annotation area can facilitate students' learning performance; c) The use of multimodal measures provided a more comprehensive assessment of participants' intuitive perception of the usability of the interface interaction design; d) Semi-structured interviews with participants provided suggestions for improving similar collaborative reading systems.

ACKNOWLEDGMENTS

This study was supported by the National Key Research and Development Program of China (No. 2020AAA0108800), the Ministry of Education of the People's Republic of China (Grant Nos. 2020AAA0108804) and the Research Funds of CCNU from the Colleges' Basic Research (Grant Nos. CCNU20ZN006).

REFERENCES

- [1] M. Zhang, G. Hou, and Y.-C. Chen, "Effects of interface layout design on mobile learning efficiency: a comparison of interface layouts for mobile learning platform," *Library Hi Tech*, vol. ahead-of-print, no. ahead-of-print, 2022, doi: 10.1108/LHT-12-2021-0431.
- [2] H. S. Christensen, "A conjoint experiment of how design features affect evaluations of participatory platforms," *Government Information Quarterly*, vol. 38, no. 1, p. 101538, 2021/01/01/ 2021, doi: <https://doi.org/10.1016/j.giq.2020.101538>.
- [3] M. N. Islam, H. Bouwman, and A. K. M. N. Islam, "Evaluating Web and Mobile User Interfaces With Semiotics: An Empirical Study," *IEEE Access*, vol. 8, pp. 84396–84414, 2020, doi: 10.1109/ACCESS.2020.2991840.
- [4] J. H. Goldberg, "Eye Movement-Based Interface Evaluation: What can and Cannot be Assessed?," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 44, no. 37, pp. 625–628, 2000/07/01 2000, doi: 10.1177/154193120004403721.
- [5] J. H. Goldberg and X. P. Kotval, "Computer interface evaluation using eye movements: methods and constructs," *International Journal of Industrial Ergonomics*, vol. 24, no. 6, pp. 631–645, 1999/10/01/ 1999, doi: [https://doi.org/10.1016/S0169-8141\(98\)00068-7](https://doi.org/10.1016/S0169-8141(98)00068-7).
- [6] J. Goldberg and X. Kotval, "Eye movement-based evaluation of the computer interface," *Advances in Occupational Ergonomics and Safety*, pp. 529–532, 01/01 1998.
- [7] L. Deng and G. Wang, "Application of EEG and Interactive Evolutionary Design Method in Cultural and Creative Product Design," (in eng), *Comput Intell Neurosci*, vol. 2019, p. 1860921, 2019, doi: 10.1155/2019/1860921.
- [8] H. Ayaz, P. A. Shewokis, S. Bunce, K. Izzetoglu, B. Willems, and B. Onaral, "Optical brain monitoring for operator training and mental workload assessment," *NeuroImage*, vol. 59, no. 1, pp. 36–47, 2012/01/02/ 2012, doi: <https://doi.org/10.1016/j.neuroimage.2011.06.023>.
- [9] T. Čegovnik, K. Stojmenova, G. Jakus, and J. Sodnik, "An analysis of the suitability of a low-cost eye tracker for assessing the cognitive load of drivers," *Applied Ergonomics*, vol. 68, pp. 1–11, 2018/04/01/ 2018, doi: <https://doi.org/10.1016/j.apergo.2017.10.011>.
- [10] C.-M. Chen and S.-H. Huang, "Web-based reading annotation system with an attention-based self-regulated learning mechanism for promoting reading performance," *British Journal of Educational Technology*, vol. 45, no. 5, pp. 959–980, 2014/09/01 2014, doi: <https://doi.org/10.1111/bjet.12119>.
- [11] X. Yang, L. Lin, P.-Y. Cheng, X. Yang, and Y. Ren, "Which EEG feedback works better for creativity performance in immersive virtual reality: The reminder or encouraging feedback?," *Computers in Human Behavior*, vol. 99, pp. 345–351, 2019/10/01/ 2019, doi: <https://doi.org/10.1016/j.chb.2019.06.002>.
- [12] J. Gonzalez-Sanchez, M. Baydogan, M. E. Chavez-Echeagaray, R. K. Atkinson, and W. Bureson, "Chapter 11 - Affect Measurement: A Roadmap Through Approaches, Technologies, and Data Analysis," in *Emotions and Affect in Human Factors and Human-Computer Interaction*, M. Jeon Ed. San Diego: Academic Press, 2017, pp. 255–288.
- [13] G. Z. Wang Xue, Zhang Lei, Wang Zhijun, "Research on Mechanisms and Strategies of Value Induction to Promote Video Learning: Analysis Based on Multimodal Data," *E-education Research*, vol. 43, no. 02, pp. 56–63, 2022, doi: 10.13811/j.cnki.eer.2022.02.008.
- [14] M. T. H. Chi et al., "Translating the ICAP Theory of Cognitive Engagement Into Practice," *Cognitive Science*, vol. 42, no. 6, pp. 1777–1832, 2018/08/01 2018, doi: <https://doi.org/10.1111/cogs.12626>.
- [15] M. T. H. Chi and R. Wylie, "The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes," *Educational Psychologist*, vol. 49, no. 4, pp. 219–243, 2014/10/02 2014, doi: 10.1080/00461520.2014.965823.
- [16] M. T. H. Chi, "Active-Constructive-Interactive: A Conceptual Framework for Differentiating Learning Activities," *Topics in Cognitive Science*, vol. 1, no. 1, pp. 73–105, 2009/01/01 2009, doi: <https://doi.org/10.1111/j.1756-8765.2008.01005.x>.
- [17] M. Benedek and C. Kaernbach, "A continuous measure of phasic electrodermal activity," *Journal of Neuroscience Methods*, vol. 190, no. 1, pp. 80–91, 2010/06/30/ 2010, doi: <https://doi.org/10.1016/j.jneumeth.2010.04.028>.
- [18] W. Boucsein, *Electrodermal Activity* // Berlin: Springer Science & Business Media, 2012.
- [19] I. Warsah, R. Morganna, M. Uyun, H. Hamengkubuwono, and M. Afandi, "The Impact of Collaborative Learning on Learners' Critical Thinking Skills," *International Journal of Instruction*, vol. 14, pp. 443–460, 04/01 2021, doi: 10.29333/iji.2021.14225a.
- [20] A. A. Gokhale, "Collaborative Learning Enhances Critical Thinking," *Journal of Technology Education*, vol. 7, no. 1, pp. 22–30, 1995, doi: <http://doi.org/10.21061/jte.v7i1.a.2>.
- [21] S. Avsec, M. Jagiello-Kowalczyk, and A. Żabicka, "Enhancing Transformative Learning and Innovation Skills Using Remote Learning for Sustainable Architecture Design," *SUSTAINABILITY*, vol. 14, no. 7, p. 3928, 2022. [Online]. Available: <https://www.mdpi.com/2071-1050/14/7/3928>.
- [22] K. Miller, S. Zyto, D. Karger, J. Yoo, and E. Mazur, "Analysis of student engagement in an online annotation system in the context of a flipped introductory physics class," *Physical Review Physics Education Research*, vol. 12, no. 2, p. 020143, 12/14/ 2016, doi: 10.1103/PhysRevPhysEduRes.12.020143.
- [23] W. Yan, X. W. Zhang, L. Deng, and Z. Y. Liu, "Research on the generality of icon sizes based on visual attention," (in English), *ENGINEERING REPORTS*, 2022, Art no. e12577, doi: 10.1002/eng2.12577.
- [24] M. Jiang, S. Liu, Q. Feng, J. Gao, and Q. Zhang, "Usability Study of the User-Interface of Intensive Care Ventilators Based on User Test and Eye-Tracking Signals," (in eng), *Medical science monitor : international medical journal of experimental and clinical research*, vol. 24, pp. 6617–6629, Sep 20 2018, doi: 10.12659/msm.909933.